SZZX-ID Bridge Program Physics

Describe Motions

江苏省苏州中学国际部
Suzhou High School – International Division

Class Outline

Title of Unit	Linear motion
Title of Lesson	Describe motion
Starter Activity	vocabulary
Direct Instructions	Teaching Strategies: Demonstration, Lecture, Collaboration, Questioning, Inquiry 1. Lecture: Definition of velocity and displacement, vector and scalar 2. Scalar and vector: collaboration: worksheet 3. Collaboration: Understand acceleration by acting it Differentiation: Content, Instructional Strategies, the Classroom, Products, Teacher 1. Content: check frequently for previous knowledge on slope pf a curve 2. Product: share additional material on slope in MB 3. Check with students on their understanding and challenges International Mindedness:
Lesson Objectives	By the end of the lesson, students will be able to: 1. Understand the definition and difference of vector and scalar 2. Differentiate distance vs. displacement, velocity vs. speed 3. Understand the physical meaning of acceleration 4. Interpret a s-t and v-t graph, and understand the meaning of the slope for each
ATLs	Identify unit ATLs and how they are being addressed in the lesson communication: Understand Command terms for DP physics
IB Learner Profile	Link the lesson to the characteristics of the IB Learner attribute Communicators: represent physic quantity with vectors, interpret graphs
Teaching Activities	1. Acceleration: act 2. Acceleration: online quiz
Plenary Session	Checking for understanding / Homework / Q & A / Reflection etc. Check feedback from students for the first topic: too much? Too little? Language?

Vocabulary

江苏省苏州中学国际部
Suzhou High School – International Division

- Scalar 标量
- Vector 矢量
- Magnitude （矢量的）大小
- Direction 方向
- Distance 距离
- Displacement 位移
- Speed 速率
- Velocity 速度
- Vertical 竖直的
- Horizontal 水平的
- Average 平均的
- Instantaneous 瞬时的
- Acceleration 加速度
- Graph 图像
- Gradient 斜率
- Slope 斜率/斜坡
- Tangent line 切线

Definitional of scalar and vector

- Scalar: a quantity that has only magnitude (numerical value)
- Vector
  - a quantity that has both magnitude and direction
  - Represented by a straight arrow. The direction of the arrow represents the direction of the vector and the length of the arrow represents the magnitude of the vector
  - Two vectors are equal **<u>only when</u>** they have the same magnitude and same direction

Be aware

Print media often indicate a variable vector quantity using bold font (e.g., $\boldsymbol{F} = m\boldsymbol{a}$).

In handwritten text, an arrow above the vector (e.g., $\vec{F} = m\vec{a}$) is common.

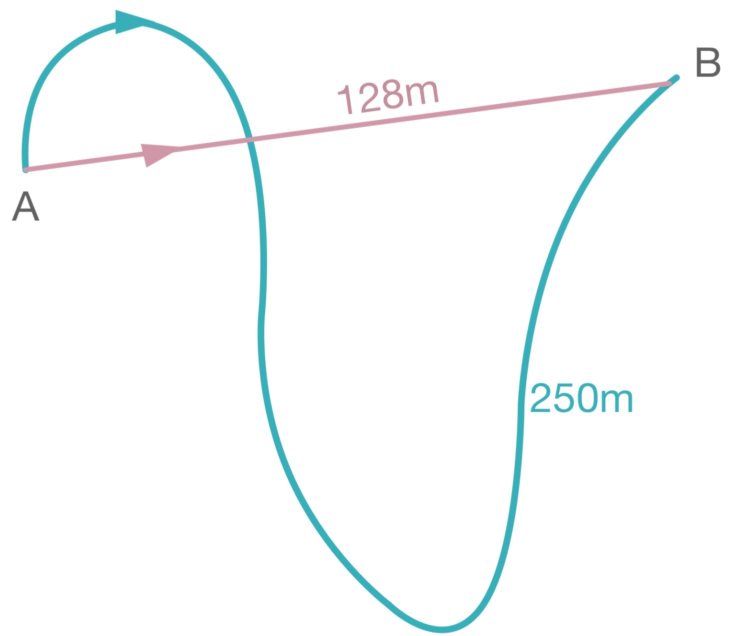
Describe motion:
Distance and Displacement

Distance, d(m)

Length of the path followed. It is a scalar.

Displacement, s(m)

The overall change in an object's position. It is a vector.



$$d_{A\text{-}B} = 250m$$

$$s_{A\text{-}B} = 128m$$

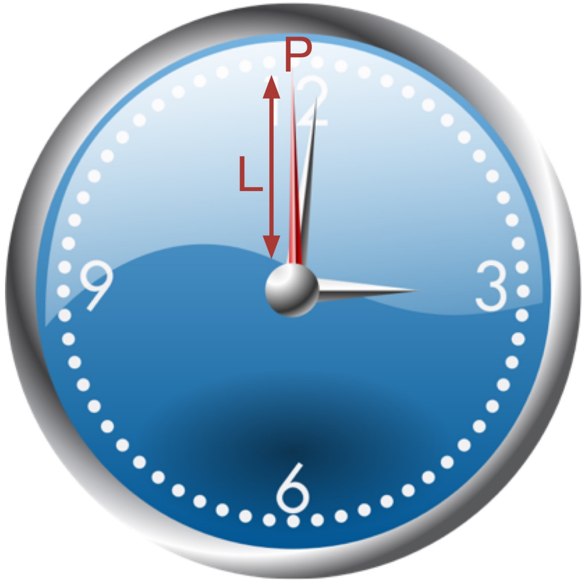
Describe motion:
Distance and Displacement

江苏省苏州中学国际部
Suzhou High School – International Division

Question

The second hand of a clock hung on a vertical wall has length *L*.
The second hand is observed at the time shown above and then again, 30 seconds later.

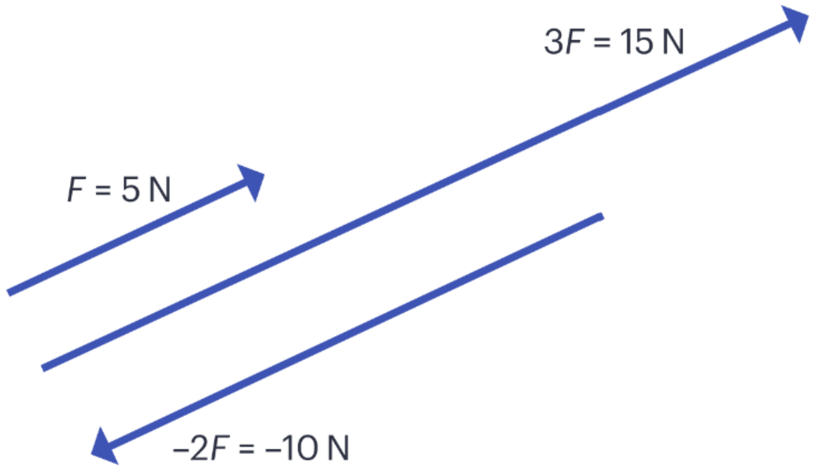
Find an expression for the displacement of, and the distance moved by, the end P of the second hand during this time interval?



Distance = πL
Displacement = 2L

Vectors in Calculation

- Multiplication of vector with a scalar
  - The magnitude of a vector can be changed by multiplying it by a scalar
  - Multiplying by a negative valued scalar gives it the opposite direction.

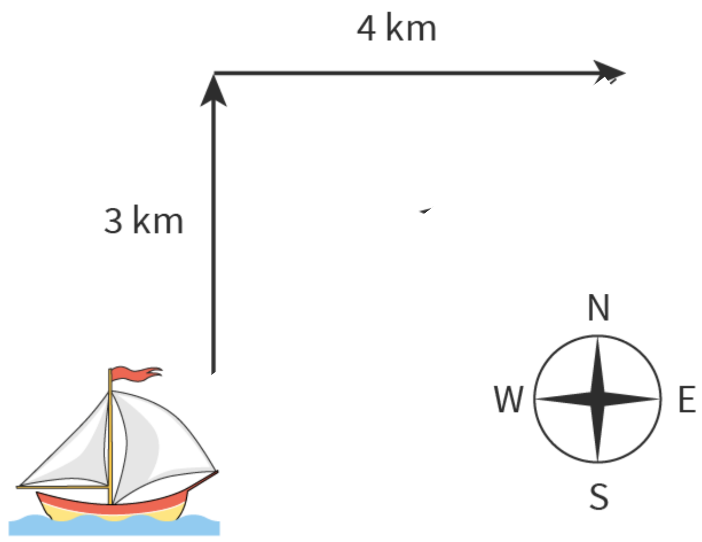


Vectors in Calculation - example

A ship travels 3 km north, and then changes direction and travels 4 km east. How far from the start is the ship?

If you draw a scale diagram of the two displacements end-to-end, the resultant can be formed by completing the third side of a triangle.

By measuring/calculating the length of the dotted line on the scale diagram, we can find the total displacement from the start. So, the ship is 5 km from the start.



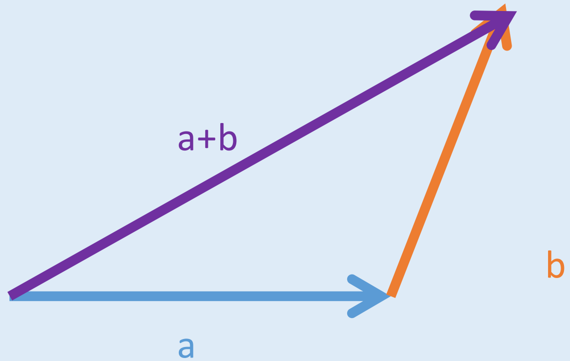
Adding vectors

江苏省苏州中学国际部
Suzhou High School – International Division

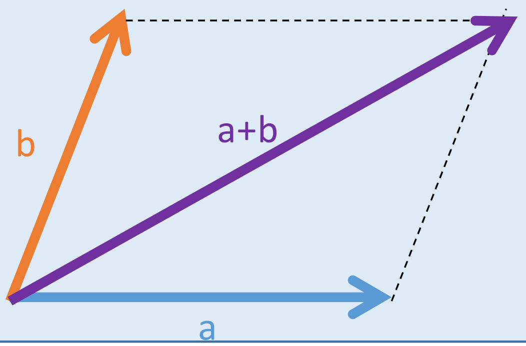
a,b are two vectors, how to find out vector (a+b)?

These are just two techniques for you to graph and calculate the sum of vectors. Essentially they are the same!

Head to tail



Parallelogram



Describe motion: Speed and velocity

Speed, s(ms^{-1})

The rate of change of distance with respect to time. It is a scalar.

$$Speed = \frac{Distance}{Time}$$

Velocity, v(ms^{-1})

The rate of change of displacement with respect to time. It is a vector.

$$Velocity = \frac{Displacement}{Time}$$

Describe motion: Speed and velocity

江苏省苏州中学国际部
Suzhou High School – International Division

An extremely fast 100 m swimmer who can complete the race in 50 seconds. What is her average speed? What is her average velocity?

Describe motion: Acceleration

江苏省苏州中学国际部
Suzhou High School – International Division

- When an object speeds up, we say that it is accelerating
- When it is slowing down, we say that it is decelerating

The graph shows two firework rockets. Imagine that you would like to buy the one that accelerates the fastest. The information given is:

- Blue rocket: 0 m/s to 12 m/s in just 4 seconds.
- Purple rocket: 0 m/s to 8 m/s in just 2 seconds.

Can you work out which one will accelerate the fastest?

Describe motion: Acceleration

- The physical quantity that we use to describe how fast velocity changes: **acceleration**
- Acceleration is the change in velocity per unit time.

$$\text{acceleration} = \frac{\text{change in velocity (m/s)}}{\text{time taken (s)}}$$

$$a = \frac{\Delta v}{\Delta t}$$

- Instead of writing “m/s per second”, we use m s^{-2} or m/s^2 for the unit of acceleration

Δ is the Greek letter delta and means 'change in...', so Δv in the equation means 'change in velocity'.

Positive and negative acceleration

A Formula 1 car is travelling at 65 m s^{-1} and is approaching a corner.
The driver applies the brakes and slows down to 35 m s^{-1} for the corner. It takes 1.2 seconds for the speed to change. What is the acceleration of the car?

$$\begin{aligned}\text{change in speed} &= 35 - 65 \\ &= -30\ \mathrm{m/s}\end{aligned}$$

$$\text{time taken} = 1.2\ \mathrm{s}$$

$$\begin{aligned}a &= \frac{-30}{1.2} \\ &= -25\ \mathrm{m/s^2}\end{aligned}$$

In this case a negative acceleration shows that the car is slowing down

Understand the sign of acceleration

江苏省苏州中学国际部
Suzhou High School – International Division

- Form groups and demonstrate the following by acting

	Velocity	Acceleration
a.	Positive	Positive
b.	Positive	Negative
c.	Positive	Zero
d.	Negative	Positive
e.	Negative	Negative
f.	Negative	Zero
g.	Zero	Positive
h.	Zero	Negative

- When v and a have the same sign, the object is accelerating
- When v and a have opposite signs, the object is decelerating
- In physics language, in both cases, we call the change of velocity acceleration